Albert Yuan

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EDUCATION

Duke University, Pratt School of Engineering **Durham, NC USA**
Master of Engineering (MEng), Electrical and Computer Engineering August 2023 - December 2024
Machine Learning/Big Data Specialization
Relevant Coursework: Random Signals and Noise, Deep Neural Nets, Probabilistic Machine Learning, Signal Detection and Estimation Theory

Duke University, Trinity College **Durham, NC USA**
Bachelor of Science (BS), Computer Science August 2019 - May 2023
GPA: 3.7 | Dean's List (2019, 2020)
Relevant Coursework: Data Structures and Algorithms, Software Design and Implementation, Advanced Algorithms, Operating Systems, Database Systems

EXPERIENCE

N1 Health
Data Ops Engineer July 2024 – Present

- **Led the design and implementation** of a centralized **feature store (dbt)**, unifying disparate data sources into a single, documented layer used by the entire data science and analytics organization; supports **200K–800K records/month** with built-in **data completeness checks** and **versioning via Apache Iceberg tables**
- **Architected and rebuilt** the end-to-end data processing pipeline (**dbt + Apache Airflow**), containerizing workloads with Docker and deploying via Amazon ECS (images in Amazon ECR); improved modularity and reliability while enabling parallelized execution and scalable throughput
- Built a **dependency-aware, change-detection data standardization system** that conditionally recomputes only affected transformations based on upstream data changes, significantly reducing redundant query execution and compute cost
- Developed internal **data ingestion and orchestration tooling** to automatically detect new client files, standardize schemas, and streamline DataOps workflows
- Led **identity resolution** for large-scale client datasets (200K–800K members) using Splink and internal tooling, resolving many-to-many identifier conflicts and improving entity-level data integrity
- Redesigned the **data quality framework (dbt tests)** with comprehensive validation (schema, completeness, query success, and custom checks), significantly improving reliability of downstream datasets

SKILLS

Coding Languages: Python, Java, C, R, Bash, HTML, CSS, Swift, Javascript
Database Systems: SQLite, Presto SQL, PostgreSQL, DBT
AWS: Athena, Glue, S3, ECS, ECR